

Syllabus for SODA 501-002
Big Social Data: Approaches and Issues (Penn State)
Spring 2026 (Regular Session)

Instruction Mode: In Person

Campus/Location: 124 Pond

Meeting Time: Thursdays, 12:00–3:00 pm

Meeting Dates (session window): 01/12/2026 – 05/01/2026

First/Last Thursday meeting: 01/15/2026 – 04/30/2026

Credits: 3

Instructor: Jared Edgerton

Email: jfe4@psu.edu

Office: Carpenter 513

Office Hours: Wednesdays 10:00 to 2:00 and by appointment

Administrative note: If you are attempting to enroll in **SODA 502**, please contact Mady Forsythe (mjf6218@psu.edu).

Course Description

SODA 501 introduces core approaches and issues in **big social data**, emphasizing the tools and practices by which social data are created, collected, curated, linked, documented, and shared. “Social” data include data about (or arising from) human interactions, including text, network, spatial, audio/video, sensor, and platform-generated data, often at scales or levels of complexity that strain conventional social-science workflows.

Course format. Each class meeting follows a consistent structure (with a short break/transition built in):

1. **Research talk (60 min):** an outside speaker presents their research and workflow.
2. **Coding lab (80 min):** a designated student **Lab Lead** runs an instructor-prepared coding demonstration and guided exercises.
3. **Paper discussion (40 min):** we discuss a social-science article that uses the week’s tool/method.

Pre-class video. For each week, students must watch a **20–40 minute video** (posted on Canvas) before class. The video introduces concepts and sets up the coding lab.

Student Learning Objectives

By the end of the course, students will be able to:

1. Identify major sources and modalities of big social data (text, network, spatial, digital trace, administrative).
2. Build end-to-end data pipelines: acquisition → storage → cleaning → linkage → documentation.

3. Implement core tooling in **Python or R** (plus SQL and Git) for scalable data work.
4. Evaluate social-science research that uses big social data, with attention to validity, ethics, and reproducibility.
5. Communicate across disciplines by engaging research speakers and translating methods across domains.

Required Tools

- A laptop capable of running Python and/or R.
- Git + a GitHub account.
- Access to Penn State Canvas for videos, announcements, and submissions.
- Optional (helpful): Docker/Podman, a cloud account (AWS/GCP/Azure), or PSU computing resources (if available).

Tooling Choice (Python vs R) and Repository Standards

Students may complete coding work in **Python or R**. You may switch languages during the semester, but each assignment must be fully reproducible in the language you choose for that assignment.

Baseline requirements for all submissions:

- A clean, reproducible project structure (at minimum: `/data_raw`, `/data_processed`, `/src`, `/docs`).
- A short **README** describing: (i) the data source and access method, (ii) how to run the code end-to-end, and (iii) what outputs are produced.
- Clear provenance notes: where the data came from, when it was collected, and any known limitations.

Recommended defaults:

- Python: `uv` or `conda` environment + `requirements.txt` (or `pyproject.toml`).
- R: `renv` lockfile.
- Optional (best practice): containerize with Docker/Podman for “one command” reproduction.

Evaluation (Grading)

Your final grade will be based on:

- **Participation and in-class engagement (10%)**
Includes Q&A with speakers, active contribution during coding lab, and substantive participation in article discussion. Participation also includes serving as an **Article Discussion Lead** at least once during the semester (graded; see below).
Article Discussion Lead (graded as part of participation). Each student will serve as **Discussion Lead** for at least one class session (more if enrollment requires). The Discussion Lead will:

1. Review the articles. Consult the useful documents folder in the Files folder of Canvas. Find the discussion guidelines document.
2. Provide 3–5 discussion questions, including at least one question about data provenance/measurement and at least one question about the computational workflow.
3. Facilitate the in-class discussion and ensure the conversation connects the paper to the week’s tool and validity threats.

- **Weekly coding labs / assignments (30%)**

Short deliverables demonstrating the week’s tool (code + brief README).

- **Student-led coding demo (20%)**

Each student will serve as **Lab Lead** for at least one class session (more if enrollment requires). The Lab Lead will:

1. Meet with the instructor **before class** (typically 30–60 minutes, scheduled during the prior week) to review the instructor-prepared demo code.
2. Rehearse the live walkthrough and confirm the demo runs on a clean environment.
3. Give a very quick highlight of material covered in the video lecture (< 5 minutes).
4. Lead the in-class coding demonstration and handle questions, with the instructor as backup.
5. Submit a short post-class note (bullet points) describing what worked, what broke, and how it was fixed.

Lab Lead and Discussion Lead assignments will be posted after Week 1 on Canvas.

- **Final project (40%)**

presentation (15%), final repo + writeup (20%), class benchmarks (5%).

Letter grades. I will use the following thresholds: A (93–100), A– (90–92.9), B+ (87–89.9), B (83–86.9), B– (80–82.9), C+ (77–79.9), C (70–76.9), D (60–69.9), F (<60).

Academic Integrity

Penn State expects students to complete all coursework in accordance with course-specific rules and university policy on academic integrity. Unless explicitly permitted, you must not submit work that is not your own, and you must not assist others in ways that violate course rules. When in doubt, ask before submitting.

Students with Disabilities

Penn State welcomes students with disabilities into the University’s educational programs. If you anticipate needing accommodations, contact Student Disability Resources (SDR) early in the semester and provide the instructor with the approved accommodation letter as soon as it is available.

Course Norms

Be professional and collaborative. This course is designed to be challenging. The goal is to build durable, transferable workflows—not just to “get it to run.”

Educational equity / bias reporting

Penn State takes great pride in fostering a diverse and inclusive environment for students, faculty, and staff. Acts of intolerance, discrimination, or harassment due to age, ancestry, color, disability, gender, gender identity, national origin, race, religious belief, sexual orientation, or veteran status are not tolerated and can be reported through Educational Equity via the Report Bias <https://equity.psu.edu/reportbias>

Counseling services

Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional well-being. The University offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. Counseling and Psychological Services at University Park (CAPS): 814-863-0395. Penn State Crisis Line (24 hours/7 days/week): 877-229-6400. Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741.

AI policy

AI tools may be used for coding support and for limited copy-editing. Acceptable uses include debugging code, explaining programming syntax, helping identify mistakes, suggesting revisions to code, and improving grammar or clarity in writing.

AI may not be used to generate the substantive written content of assignments. Interpretation, explanation, argument, and written analysis must come from the student. In other words, AI may help you write code more effectively and polish your prose, but it may not do your thinking for you.

Students remain fully responsible for all submitted work. Any AI-assisted work must comply with Penn State's academic integrity expectations.

Important Dates

- February 12: Submit two abstracts on Canvas and review the assigned group.
- March 5: Submit a one page write up to Canvas and review the assigned group.
- April 16: Submit a two page write up to Canvas and review the assigned group.
- April 30: Final presentation.
- May 7: Submit final paper.

Course Schedule (Spring 2026 — Thursdays)

Key dates: Spring Break week is Mar 8–14 (no class Thu Mar 12). Instructor is traveling Mar 26. You can attend session 1 if you want the material but it is optional. Group presentations are Thu

Apr 30. Final paper is due Fri May 1. **Readings/videos:** Posted on Canvas; the “application paper(s)” is discussed during the last 30 minutes.

Week 01: January 15, 2026 — Course Overview & What is “Big Social Data”?

- **Coding lab:** Course toolchain setup (Python/R environment, Git/GitHub, repo structure).
- **Application paper:** Lazer et al. (2009) “Computational Social Science” (Science) *or* instructor-selected modern replication.
- **Pre-class video:** Workflow overview + reproducible project scaffolding.

Week 02: January 22, 2026 — Reproducibility: Git, Containers, and Research Artifacts

- **Coding lab:** Git branching; pull requests; Docker/containers; environment capture; data documentation.
- **Application paper:** Heath, Davidson, et al. (2023) The Journal of Finance
- **Application paper:** Holzmeister, et al. (2025) Nature Human Behaviour
- **Pre-class video:** Reproducible pipelines end-to-end.

Week 03: January 29, 2026 — The Web as Data (Scraping & HTML)

- **Coding lab:** Scraping with BeautifulSoup/requests (Python) or rvest (R); respectful scraping; pagination.
- **Application paper:** Brown et al. (2025) Big Data and Society.
- **Application paper:** Mahdavi, P. (2017) PSRM
- **Pre-class video:** Web scrapping overview.

Week 04: February 5, 2026 — Text as Data Pipelines

- **Coding lab:** Tokenization; embeddings; topic models; basic transformer workflow (lightweight).
- **Application paper:** Alsarra et al. (2025) Computational and Mathematical Organization Theory
- **Application paper:** Burnham et al. (2025) Political Analysis
- **Pre-class video:** Practical text pipeline architecture.

Week 05: February 12, 2026 — RCTs and Large Experiments (Design + Implementation at Scale)

- **Coding lab:** Randomization checks; blocking/stratification; power; clustered designs; analysis of ATE; logging and reproducibility for large experiments.
- **Application paper:** Instructor-selected large-scale field or platform experiment (to be posted on Canvas).
- **Application paper:** Bond et al. (2012) Nature; Allcott et al. (2020) AER.
- **Pre-class video:** RCT design basics + practical issues at scale (interference, spillovers, compliance, clustered assignment).
- Submit two research abstract ideas for your group paper to Canvas.
 - Group 1 will review groups 2’s ideas.
 - Group 2 will review groups 3’s ideas.

- Group 3 will review groups 4’s ideas.
- Group 4 will review groups 1’s ideas.

Week 06: February 19, 2026 — Time Series Analyses for Social Data

- **Coding lab:** Time-indexed data cleaning; temporal splits; forecasting vs causal timing; ARIMA/ETS baselines; evaluation under temporal cross-validation.
- **Application paper:** Chen et al. (2022) PA; Park and Yamauchi (2022) PA.
- **Pre-class video:** Time series workflow design + leakage pitfalls (temporal validation, rolling windows, backtesting).

Week 07: February 26, 2026 — LLM-Assisted Data Extraction (Human-in-the-Loop)

- **Coding lab:** Structured extraction from messy text using LLMs; prompt design for schemas; batch processing; uncertainty checks; human validation/spot-audits; documenting failure modes.
- **Application paper:** Gilardi et al. (2023) PNAS
- **Application paper:** Heseltine, M. & Bernhard Clemm von Hohenberg. (2024) Research & Politics
- **Pre-class video:** LLM extraction patterns + evaluation (precision/recall, auditing).

Week 08: March 5, 2026 — APIs, Rate Limits, and Data Collection at Scale

- **Coding lab:** API auth; pagination; retries; logging; storing raw JSON.
- **Application paper:** Bail et al. (2018) PNAS
- **Application paper:** Barbeta, P. (2015) Political Analysis
- **Pre-class video:** API design patterns + data provenance.
- Revise your group presentation ideas. Identify a project you plan to coauthor, a data source, a research question, and lay out an analytic overview (1 page)
 - Group 4 will review groups 2’s ideas.
 - Group 3 will review groups 1’s ideas.
 - Group 2 will review groups 4’s ideas.
 - Group 1 will review groups 3’s ideas.

March 12, 2026 — No Class (Spring Break Week)

Week 09: March 19, 2026 — Data Quality, Labeling, and Validation

- **Coding lab:** Validation checks; inter-coder reliability; basic monitoring; documenting limitations.
- **Application paper:** Lazer et al. (2014) Science
- **Application paper:** Parkinson, S. (2024) APSR
- **Application paper:** Sales et al. (2022) Applied Geography
- **Pre-class video:** Validity threats in big social data.

Week 10: March 26, 2026 — Spatial Data, GIS, and Remote Sensing *Instructor is traveling attendance to the speaker is required. You may attend sessions 1 for the course material*

- **Coding lab:** sf/GeoPandas; joins; rasters; mapping; basic spatial features.
- **Application paper:** Harari, M. (2020) AER
- **Application paper:** Panagopoulos et al. (2023) Land
- **Pre-class video:** Spatial data formats + coordinate systems.

Week 11: April 2, 2026 — Databases & SQL for Social Data

- **Coding lab:** SQL basics; schema design; indexes; joins; using duckdb/sqlite; query from Python/R.
- **Application paper:** Baumgartner et al. (2020) Proceedings of the international AAAI conference on web and social media.
- **Application paper:** Green et al. (2025) APSR.
- **Pre-class video:** Relational thinking + query optimization basics.

Week 12: April 9, 2026 — Network Data & Graph Workflows

- **Coding lab:** Graph construction; centrality; community detection; scalability tips.
- **Application paper:** Minhas, S & P. Hoff (2025) Political Analysis
- **Application paper:** Olivella et al. (2022) JAS
- **Pre-class video:** Network representation choices and pitfalls.

Week 13: April 16, 2026 — Record Linkage and Entity Resolution

- **Coding lab:** String distances; blocking; probabilistic linkage; evaluation.
- **Application paper:** Ornstein, J. (2025) Political Analysis
- **Application paper:** Nyseth Nzitatira et al. (2024) ASR
- **Application paper:** Nyseth Nzitatira et al. (2022) JPR (optional)
- **Pre-class video:** Linkage as measurement + error tradeoffs.
- **Milestone:** Final project proposal due.
- **Revise your group presentation ideas.** Write your introduction and find ten articles from top field journals to center your literature review and theory. This should be two pages total.
 - Group 4 will review groups 1's ideas.
 - Group 3 will review groups 4's ideas.
 - Group 2 will review groups 3's ideas.
 - Group 1 will review groups 2's ideas.

Week 14: April 23, 2026 — Surveys, Platforms, and Crowdsourcing

- **Coding lab:** Survey exports; cleaning; codebooks; labeling; quality checks.
- **Application paper:** Berinsky et al. (2012) Political Analysis
- **Application paper:** Bisbee (2024) Political Analysis
- **Pre-class video:** Data quality, missingness, and measurement in platform data.

Week 15: April 30, 2026 — Student Project Presentations & Wrap-Up

- **Deliverable:** 20 minute presentation + demo of repo/pipeline.

Final Project

Goal: Build a reproducible pipeline that produces (or substantially augments) a “big social data” asset relevant to your research interests (text, network, spatial, platform trace, administrative, etc.).

Minimum components:

- You will coauthor the paper with your group.
- A clearly documented data source (provenance) and collection strategy.
- A reproducible pipeline (scripts/notebooks + README) that runs end-to-end.
- Basic validation checks (at least two) and a short writeup describing limitations.
- A shareable research artifact: a GitHub repo with instructions to reproduce outputs.

Milestones:

- **Presentation:** Week 15.
- **Final submission:** Due **Friday, May 8, 2026**, 5:00 pm ET (end of finals week).